Ex. No.:1	Building E-commerce Chatbot	URK23CS1167
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Aim:

To build an AI-powered E-Commerce Chatbot using Retrieval-Augmented Generation (RAG) with Ollama.

Problem Statement:

Design and develop a modular Retrieval-Augmented Generation (RAG) based e-commerce chatbot using Ollama. The chatbot should answer FAQs, recommend products, answer product-specific questions, summarize customer reviews, and intelligently route user queries using an LLM-based intent detector.

Software Requirements:

Software	Version
Python	3.13.18
Ollama	0.6.2
VS Code	1.133.0
Pandas	3.0.5
Sentence Transformers	5.7.0
Flask	3.1.3
LLM Model	Llama 3.2 (3B) via Ollama

Dataset Description:

Dataset	Purpose
Products	products.csv – 20 products (ID, name, brand, category, price, description). Used for TF-IDF product recommendation and product-specific question answering (RAG).
FAQs	faqs.csv – 16 question/answer pairs on orders, delivery, returns, refunds, payment and warranty. Used for semantic FAQ search with embeddings and cosine similarity.
Reviews	reviews.csv – 37 customer reviews with ratings. Used for sentiment analysis, review summarization and buying suggestions.

System Architecture:

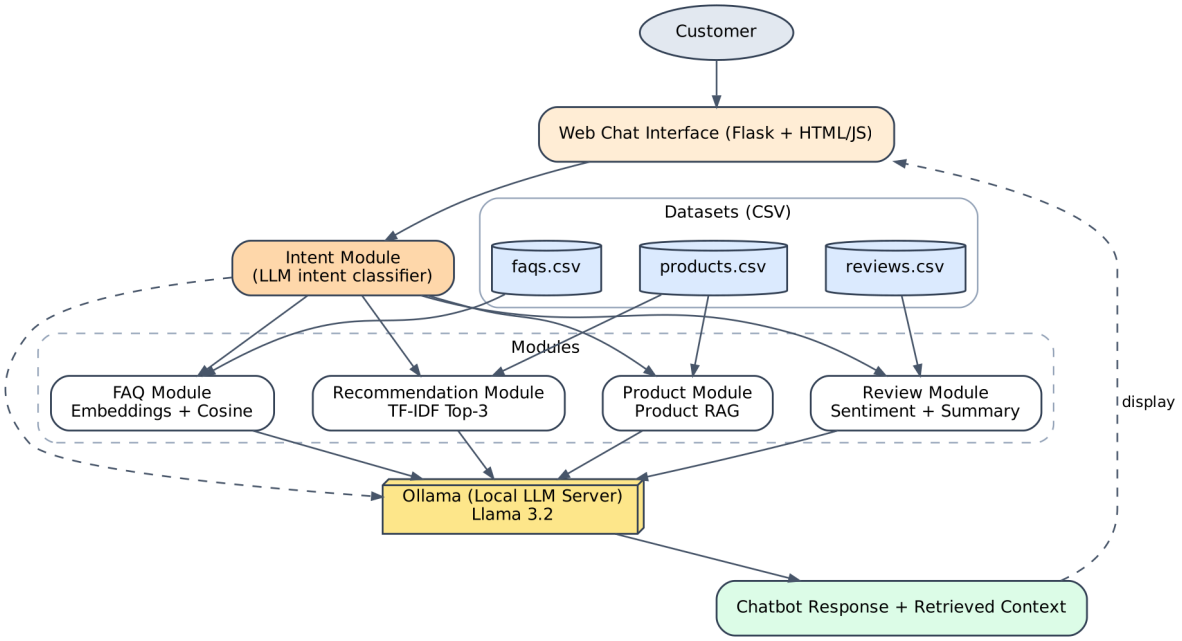


Fig. 1.1 System architecture

Module Design:

Module	Responsibility
FAQ Module	Converts FAQs into embeddings (Sentence Transformers) and finds the closest FAQ to the user query using cosine similarity. The LLM rewrites the matched answer politely.
Recommendation Module	Detects category and budget from the query, finds the top-3 similar products using TF-IDF and cosine similarity, and asks Ollama to recommend the best one.
Product Module	Retrieves the product the user is asking about and answers the question using only that product's structured information (RAG).
Review Module	Labels each review Positive/Negative/Neutral using a sentiment lexicon and generates a summary, pros, cons and a buying suggestion with the LLM.
Intent Module	Uses the LLM to classify the query as FAQ, RECOMMENDATION, PRODUCT_QA or REVIEW, with a keyword fallback, and routes the query to the matching module.
Chatbot	Integrates all modules and provides continuous chat interaction through the Flask web interface, showing the detected intent and retrieved context.

Implementation:

- GitHub Link - https://github.com/MONICK10/GPT4-Exp1-ECommerce-Chatbot
- LinkedIn Post Link - <add LinkedIn post link in config.json>
- Run: `python app.py` → `http://localhost:5001`

Output Screenshots:

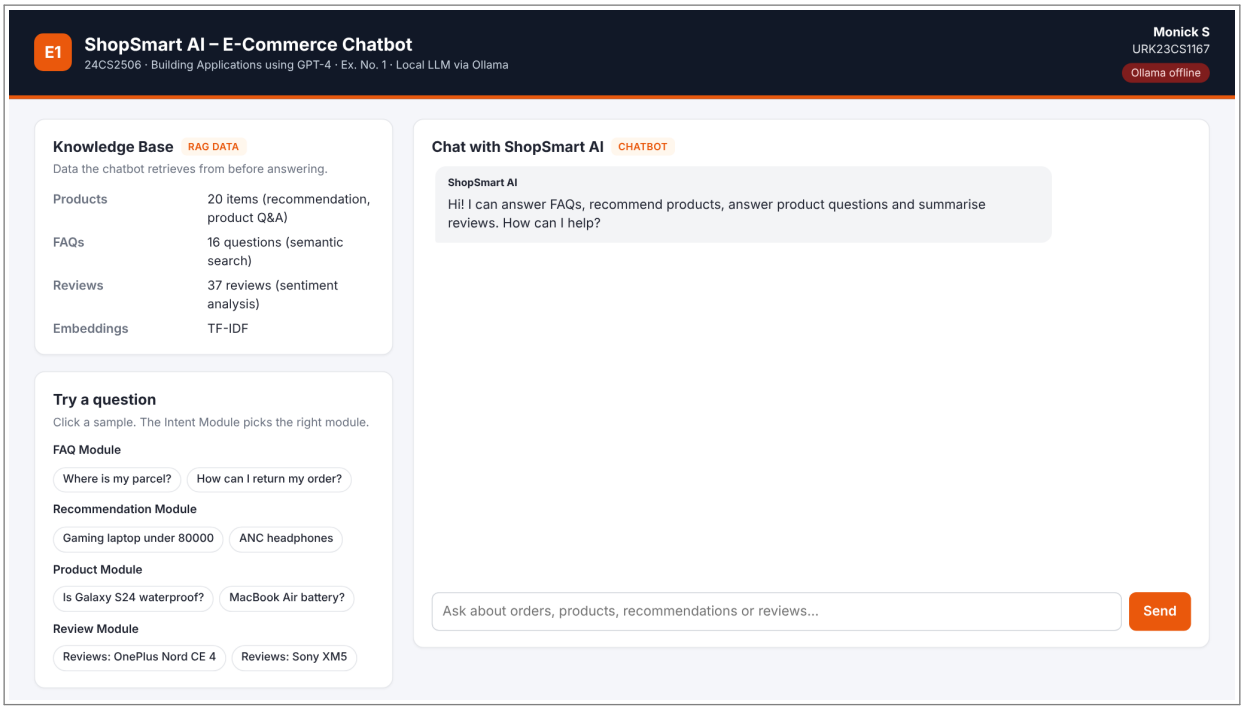


Fig. 1.2 Home Screen

[Draft] The AI output screenshots are added automatically when you run python tools/capture.py 1 on your laptop with Ollama.

Result:

Thus successfully built a local chatbot system that can answer product-related questions from a dataset, recommend products based on similarity, analyze sentiment from user/product reviews, and simulate FAQ responses using Ollama.